Google Ads API Tool — Design Document

Internal campaign scheduling and reporting tool
CÔNG TY TNHH VẬN TẢI ĐẠI LỘ XANH (Dai Lo Xanh Transport Co., Ltd.)

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Website	https://breach-hang-ballot-occurring.trycloudflare.com/xe-cau (public home page; privacy policy at /privacy)
Contact	vantaidailoxanh2026@gmail.com
Google Cloud project	686766750544 (owns the OAuth client used by the tool)
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1. Summary

We are a small transport and crane-rental company in Phu Tho province, Vietnam. We advertise our own services on Google Search from a single Google Ads account that we own. This document describes an internal back-office web application that (1) reads the performance of our own campaigns and (2) switches our existing Search campaigns on and off on the planned dates. The tool is used only by our own staff. It is not offered to other advertisers and it does not manage anyone else's account.

2. Business model and audience

- **Business model:** we sell crane-truck services, calibration test weights and freight directly to local businesses (factories, scale operators, contractors). Customers contact us by phone, Zalo or the web form. We do not sell advertising or software.
- **Advertising:** Google Search campaigns with a small monthly budget (about VND 5.3 million). A fixed campaign targets our home province; short campaigns target a destination province for a few days when our trucks are returning from there.
- **Audience of the tool:** the owner and two or three authorized employees. Access requires an individual login with role-based permissions. There are no external users.

3. What the tool does

Feature	Google Ads API usage	Frequency	Status
Connection test	GoogleAdsService.Search on the *customer* resource (name, currency)	On demand	In development
Campaign list and linking	GoogleAdsService.Search on *campaign* (id, name, status, daily budget). Read only.	Hourly	In development
Performance reporting	GoogleAdsService.Search on *campaign*: impressions, clicks, cost, conversions, phone calls for the campaign's date range. Read only.	Hourly	In development (entered manually today)
Scheduled enable / pause	CampaignService.MutateCampaigns, update mask *status* only (ENABLED / PAUSED)	Only on the planned start and end dates	In development, dry-run by default

Already in production without the API: a local campaign planner (province, dates, daily budget), a daily 07:30 reminder to start or stop campaigns, manual entry of results, and a readiness checklist for tracking and connections (Figures 1 and 2).

What the tool does not do

- It does not create campaigns, ad groups, ads or keywords, and does not change budgets, bids, targeting or creatives.
- It does not access any account other than our own, does not use a manager (MCC) account to serve third parties, and does not resell or expose Google Ads data.
- It does not use planning services (Keyword Plan, Audience Insights, Reach Plan), billing, account creation or user management.

4. Supported campaign types

Search campaigns only (including phone-call assets). The tool reads and switches the status of existing campaigns; Display, Video, Shopping, Performance Max, App and Demand Gen campaigns are not used or supported.

5. Architecture

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Staff browser  --HTTPS (Cloudflare)-->  Web admin (Python / FastAPI)  <--->  PostgreSQL
                                        |                        (campaigns, metrics,
                                        |                         audit log, encrypted keys)
Background scheduler (Python thread, every 10 min)
    |-- daily reminder ------------------------------> Telegram (staff notification)
    |-- hourly: OAuth token refresh, then REST calls --> Google Ads API
              googleAds:search   (read)
              campaigns:mutate   (status only, only when the owner has enabled live mode)
```

6. Authentication and credential handling

- OAuth 2.0 with offline access, scope *https://www.googleapis.com/auth/adwords*. The owner of the Google Ads account authorizes once; the OAuth client belongs to Google Cloud project 686766750544.
- The refresh token and client secret are encrypted (AES via Fernet) before being stored in the database. Access tokens (one hour) are kept in memory only.
- Secrets are never shown again in the interface; the settings form shows only "saved (…last 4 characters)" (Figure 2).
- Calls use Google's current access model, in which access levels belong to the Cloud project. No developer-token header is sent.

7. API volume

With at most 10 linked campaigns the tool makes about 11 read requests per hour (264 per day) and fewer than 20 status changes per day, i.e. well under 400 operations per day, far below the Basic access limit of 15,000. Requests are retried with back-off on HTTP 429 and 5xx errors (1 s, then 3 s) and never in a tight loop.

8. Safety controls

- **Dry-run by default:** until the owner explicitly turns on "live" mode with a confirmation, the tool only records what it *would* change and sends no change to Google.
- At most 10 status changes per run; a campaign whose end date has passed is never enabled; budgets are never modified.
- Every action (real or dry-run) is written to an audit table with time, campaign, action and result, and is visible to the owner.
- If the connection test fails the tool stops and reports the problem in plain language; nothing is changed.
- Rate-limit and authorization errors are classified and shown to the owner with the action to take.

9. Data handling and privacy

- Stored locally per campaign: Google campaign id, name, status, impressions, clicks, phone calls and cost.

No personal data of people who saw or clicked ads is received or stored.

- Data is used only to show results to our staff and to schedule our own campaigns. It is not sold, shared or used for any other purpose.
- The tool follows the Google Ads API Terms and the Google API Services User Data Policy including the Limited Use requirements. Our privacy policy is public at /privacy on the website above.

10. User interface

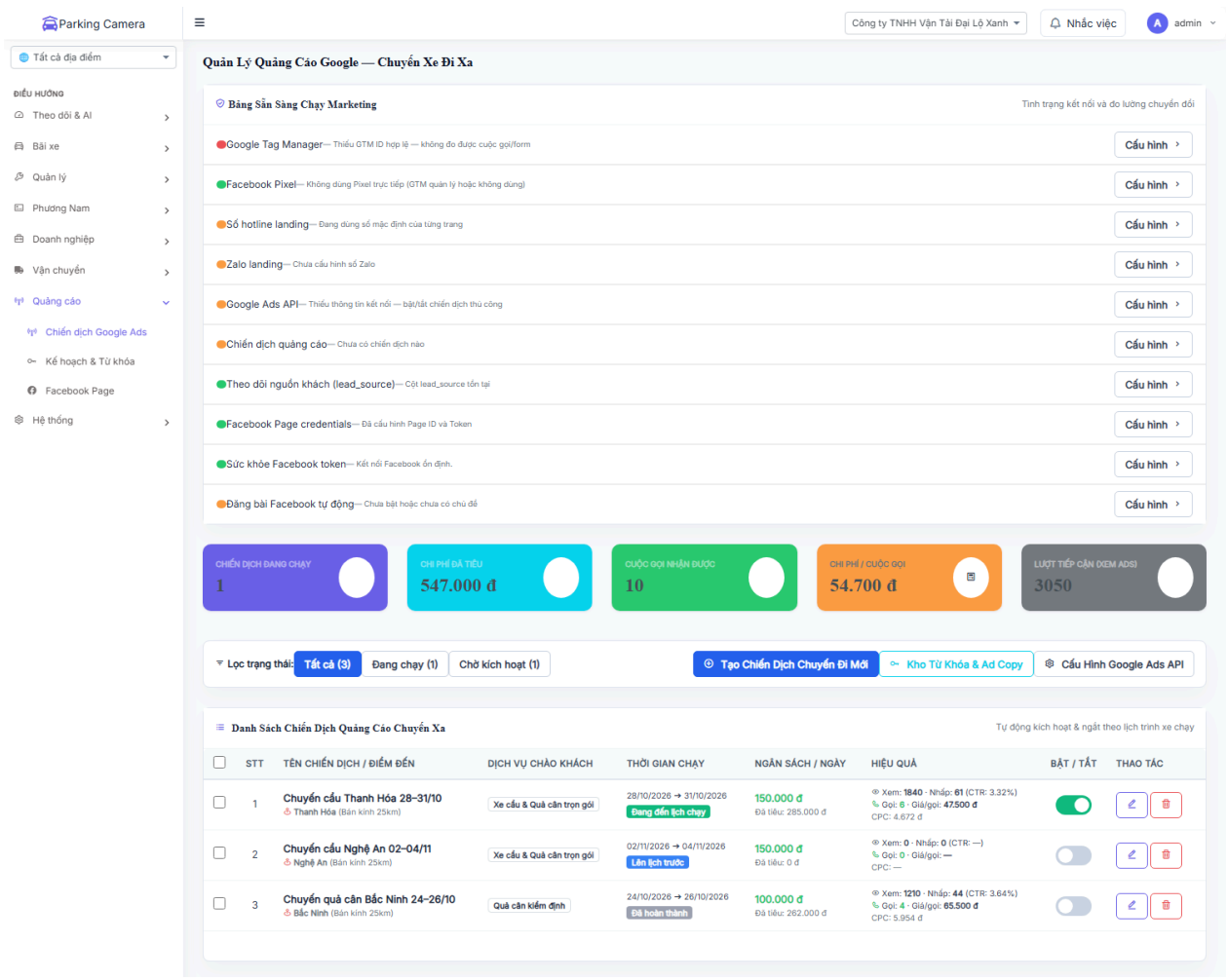


Figure 1. Campaign dashboard (current version, sample data): readiness checklist, summary figures, campaign list with start/ end dates, budget, results and the on/off switch. Results are typed in today; with the API they are filled in automatically.

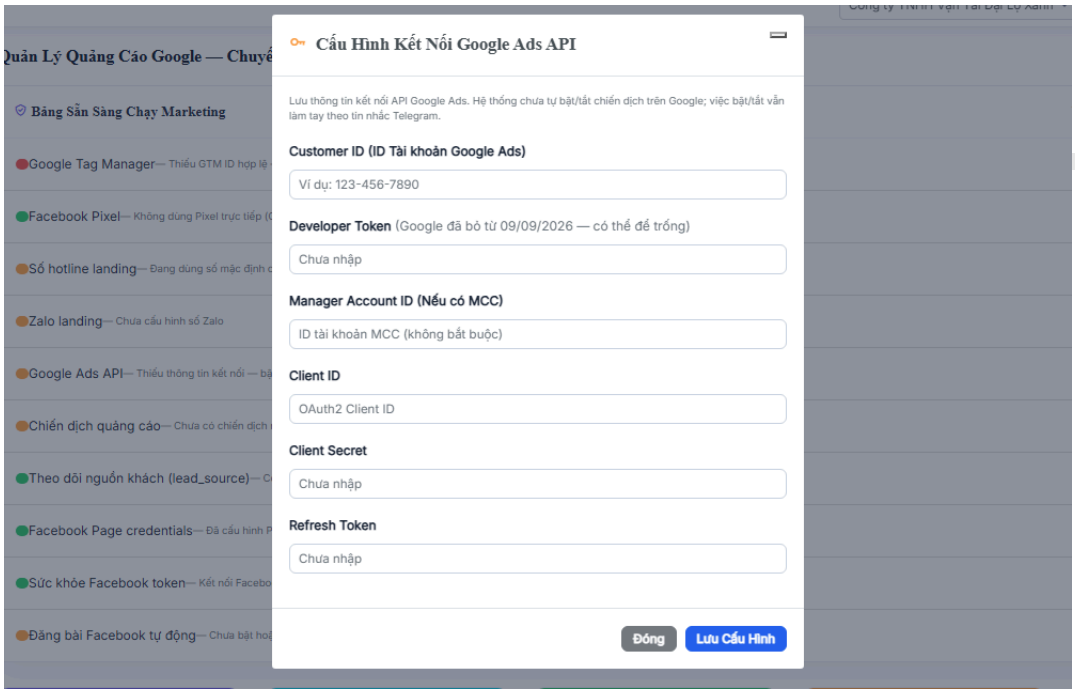


Figure 2. Connection settings for the Google Ads account. Secret fields are masked once saved.

11. Status and plan

Phase	Scope	State
1	Local planner, daily reminders, manual results entry, readiness checklist (no API calls)	In production
2	OAuth connection, connection test, read-only reporting and campaign linking, dry-run status changes	In development; first run after access approval
3	Live enable / pause, only after at least three days of dry-run results match the changes the owner makes by hand	Planned

All screenshots use sample data. The company name above is the legal name used on the OAuth consent screen and on the website.